

HÜSEYİN YURTSEVEN

Senior Engineering & Science Leader | Machine Learning & Personalization

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EXECUTIVE SUMMARY

Accomplished engineering and science executive with 20+ years of software development experience and 10+ years leading high-performance engineering and science teams at scale. Recognized leader in AI/ML and Data Science organizations, with proven expertise building and scaling cross-functional teams of Applied Scientists, Machine Learning Engineers, and Data Scientists. Track record of delivering measurable business impact through ML-powered personalization, recommendation systems, and data-driven decision-making at companies including Amazon, Alibaba Group, and Delivery Hero.

Deeply experienced in cross-functional leadership across engineering and science functions — building and leading data science teams since 2015 and partnering closely with Applied Science, Machine Learning Engineering, and Data Science teams to translate research into production systems at scale. Repeatedly transformed functional data science organizations into cross-functional data product setups (Delivery Hero, Trendyol). Proven ability to deliver results across organizations of all sizes, from high-growth startups to large-scale enterprises, by building cross-functional organizations with clear independent value streams, strategic domain-driven design approaches, and empowered team topologies that optimize for autonomy, ownership, and fast flow of delivery.

Core Competencies: - AI/ML & Data Science Leadership - Cross-Functional Team Management (Applied Science, ML Engineering, Software Development) - Machine Learning Systems at Scale (Personalization, Ranking, Recommendation) - Distributed Systems & Microservices Architecture - Organizational Design & Team Topologies - Product Development & Strategic Planning

PROFESSIONAL EXPERIENCE

Amazon / Amazon Music

Senior Software Development Manager, Amazon Music Machine Learning & Managing Director, Snackable OÜ (Estonia) *January 2023 - Present | Berlin, Germany*

Lead 3 globally distributed cross-functional teams of 30+ Applied Scientists, Machine Learning Engineers, and Software Development Engineers within Amazon Music Personalization organization across Berlin and Tallinn.

Key Responsibilities & Achievements: - Direct leadership of 3 teams across 2 regions: Customer Intelligence (Tallinn), Content Expertise (Berlin), and Candidate Intelligence (Berlin) - Own the Domain Intelligence platform — the core "Personalization as a Service" layer powering customer understanding, content expertise, and intelligent candidate retrieval for all Amazon Music experiences - Drive strategic technical initiatives in personalization algorithms, ranking systems, and customer behavior modeling - Architect and implement scalable ML systems serving millions of Amazon Music customers globally - Collaborate with product, science, and engineering leadership to define roadmap and technical strategy - Establish engineering and science excellence practices, mentoring senior engineers and applied scientists

Technical Focus Areas: - Customer Understanding & Behavioral Modeling (Customer Intelligence — Tallinn) - Content Representations & Music Expert Models (Content Expertise — Berlin) - Candidate Retrieval, Ensembling & First-Pass Ranking (Candidate Intelligence — Berlin)

Trendyol / Alibaba Group

Head of Engineering & Managing Director, Germany Office *December 2021 - January 2023 | Istanbul, Turkey & Berlin, Germany*

Led engineering organization for local commerce (Food and Groceries Delivery) vertical, managing managers of managers and overseeing 17+ teams across multiple domains including logistics, assortment, and store operations. Concurrently served as Managing Director of Trendyol's Germany office, overseeing local operations and market expansion.

Key Responsibilities & Achievements: - Established organizational structure and team topologies for 17+ engineering teams, optimizing for autonomy and delivery velocity - Served on tribe leadership group, shaping company-wide technical strategy and architectural decisions - Led Trendyol's Germany office as Managing Director, driving market presence, operational setup, and local team management - Partnered with data engineering and data science teams to build data-driven decision-making capabilities - Transformed the functional data science organization into a cross-functional data product setup, embedding data scientists directly into domain teams - Ensured consistent delivery of high-quality applications across logistics, assortment, and store operations domains - Implemented resource allocation framework ensuring predictable delivery across multiple concurrent projects

Organizational Impact: - Introduced OKR (Objectives and Key Results) framework across engineering organization - Established Executive Action Team for rapid decision-making on critical technical initiatives - Implemented RFC/ADR process, significantly improving architectural decision quality and documentation - Transformed organization to outcome-driven, empowered team model - Enhanced cross-team collaboration and reduced delivery cycle times

Technologies: Python, FastAPI, Golang, gRPC, React, scikit-learn, TensorFlow, PyTorch, Microservices, MongoDB, PostgreSQL, Elasticsearch, Nginx, Drone CI/CD, AWS, GCP

Delivery Hero SE

Director, Software Engineering *July 2020 - December 2021 | Berlin, Germany*

Led Q-Commerce engineering tribe, managing managers of managers, senior data science managers, and engineering teams. Drove technical strategy and organizational excellence for rapid-growth business unit.

Key Responsibilities & Achievements: - Directed engineering organization including platform teams, data science teams, and product engineering teams - Transitioned the functional data science organization into a cross-functional data product setup, embedding data scientists into domain teams with end-to-end product ownership - Served on tribe leadership committee, establishing technical vision and engineering standards - Architected and oversaw delivery of multiple mission-critical applications and platform services - Designed team topologies optimized for business domain alignment and technical dependencies - Managed technical interviews and performance management for senior engineering and data science talent

Technical & Business Impact: - Launched comprehensive Product Information Management (PIM) system, improving catalog quality and operational efficiency - Redesigned team structure resulting in measurable improvements in delivery velocity and quality metrics - Established metrics-driven culture with data-informed decision-making frameworks - Implemented OKR methodology across tribe, aligning engineering efforts with business outcomes - Developed multi-criteria ranking solution addressing data discrepancy challenges and improving search relevance

Technologies: Python, FastAPI, Golang, gRPC, React, scikit-learn, TensorFlow, PyTorch, Microservices, MongoDB, PostgreSQL, Elasticsearch, Varnish, Nginx, Drone CI/CD, TDD, AWS, GCP, Pub/Sub

Kariyer.net

Chief Technology Officer *February 2020 - July 2020 | Istanbul, Turkey*

Led technology organization of Turkey's premier job board and ATS platforms (Kariyer.net and İşin Olsun), leading the organization as an R&D company and managing 80+ engineers, data scientists, DevOps engineers, and QA professionals.

Key Responsibilities & Achievements: - Directed all engineering, R&D, data science, DevOps, and QA functions across multiple product lines - Established software architecture standards and scalability principles for high-traffic platforms - Implemented OKR and Agile methodologies (Scrum, Kanban) across

engineering organization - Led recruitment, training, and performance management for technical teams - Ensured predictable delivery of features and applications through resource planning and allocation

Technologies: .NET, .NET Core, Python, Sanic, React, Next.js, Vue.js, Nuxt.js, JWT, scikit-learn, TensorFlow, PyTorch, Microservices, MongoDB, SQL Server, PostgreSQL, Elasticsearch, Varnish, Nginx, Jenkins CI/CD, TDD, Azure

BluTV

Chief Technology Officer *April 2018 - February 2020 | Istanbul, Turkey*

Led technology vision and execution for Turkey's leading OTT streaming platform, managing 40+ engineers, data scientists, product owners, DevOps engineers, and QA professionals.

Key Responsibilities & Achievements: - Directed end-to-end technology development for comprehensive OTT platform serving millions of users - Built in-house capabilities across video delivery, DRM, recommendation systems, and content management - Established engineering culture focused on quality, scalability, and innovation - Managed cross-functional collaboration between engineering, product, and business stakeholders - Implemented Agile methodologies and modern software development practices

Technical Innovations: - Developed proprietary video streaming infrastructure supporting HLS, MPEG-DASH, and Smooth Streaming - Implemented multi-DRM solution (PlayReady, FairPlay, Widevine) for content protection - Built in-house data science team and ML-powered recommendation and personalization systems, driving content discovery and engagement across the platform - Architected microservices-based platform for scalability and reliability

Technologies: Python, Sanic, Tornado, React, React Native, Next.js, Vue.js, Nuxt.js, JWT, scikit-learn, Microservices, MongoDB, PostgreSQL, Elasticsearch, Varnish, Nginx, HLS, MPEG-DASH, PlayReady, FairPlay, Widevine, Video.js, Shaka Player, ExoPlayer, Jenkins CI/CD, TDD, AWS

Dogan TV

Director of Software Development and Technical Innovation *September 2015 - April 2018 | Istanbul, Turkey*

Led 70+ person engineering and data science organization across multiple media properties and platforms, driving technical innovation, applied research, and architectural excellence.

Key Responsibilities & Achievements: - Managed managers of managers, establishing multi-level leadership structure - Drove Dogan TV's transformation into an R&D company, establishing data science and applied research as core functions - Built and led the company's first data science team - Established the company's first Hadoop cluster, creating the data platform foundation for ML across all properties - Architected and deployed Hadoop-based ML systems for recommendation, churn prediction (powering subscriber retention for pay-TV services), and audience segmentation - Built ML-based image matching and similarity capabilities into Blupoint CMS - Directed development of headless CMS (Blupoint CMS), eCRM, DRM systems, and custom CDN solutions - Processed and analyzed clickstream and video event data from 40+ platforms - Implemented full DRM license servers (Widevine, FairPlay, PlayReady)

Platform Portfolio: blutv.com.tr, netd.com, kanald.com.tr, cnnturk.com, dsmart.com.tr, radyonom.com, dreamtv.com.tr

Technologies: Python, Tornado, Vue.js, Nuxt.js, JWT, scikit-learn, Microservices, MongoDB, Elasticsearch, Varnish, Nginx, Hadoop, HLS, MPEG-DASH, Smooth Streaming, DRM technologies

Software Development and Technical Innovation Manager *January 2014 - September 2015*

Managed Dogan TV Software Development Team, leading application engineers and web developers for netd.com, kanald.com.tr, and cnnturk.com.

Key Achievements: - Built multi-site flexible CMS system adopted across all Dogan TV properties - Implemented technology stack: MongoDB, Elasticsearch, Redis, ASP.NET MVC, Web API, OData, BackboneJS, AngularJS - Built Dogan TV's first recommendation system, developing Python-based ML

algorithms adopted across the company's properties - Created HLS-capable Flash and HTML5 video players with VAST 3 advertising support

netd.com - Software Development Manager *August 2012 - January 2014*

Managed netd.com software development team, delivering high-quality applications and establishing architectural standards.

cnnturk.com - Software Development Manager *March 2009 - August 2012*

Led cnnturk.com engineering team, managing weekly releases, capacity planning, and cross-functional coordination between business and engineering.

Key Contributions: - Developed mobile applications for iPhone, iPad, Android, Nokia Ovi, and Samsung Bada platforms - Conducted capacity planning ensuring system stability during high-traffic events - Established release engineering processes and quality standards

cnnturk.com - Senior Software Developer *March 2008 - March 2009*

Promoted to management within one year. Developed cnnturk.com CMS system and maintained internet properties.

VERUS BILISIM

Software Development Team Leader *October 2005 - March 2008 | Istanbul, Turkey*

Led development teams delivering diverse applications including congress management systems, shipping/cargo management software, ERP solutions, and CMS platforms.

Technologies: Visual C#, ASP.NET, SQL Server, PostgreSQL, MySQL, SQL Server Reporting Services, FastReport, Crystal Reports

HYBK BILISIM

Web Developer *January 2002 - June 2005 | Istanbul, Turkey*

Developed web applications and solutions during academic studies.

PUBLICATIONS

A Unified Recommendation Model for Features Summarization Amazon Science <https://www.amazon.science/publications/a-unified-recommendation-model-for-features-summarization>

TECHNICAL SKILLS

Leadership & Management: Coaching, Mentoring, Team Building, Organizational Design, Cross-Functional Leadership, Strategic Planning, OKR Implementation, Agile/Scrum/Kanban Methodologies

Architecture & Systems: Distributed Systems, Microservices Architecture, RESTful APIs, Event-Driven Architecture, Scalability Design, System Design, Cloud Architecture (AWS, GCP, Azure)

Programming & Frameworks: Python (FastAPI, Sanic, Tornado, Django, scikit-learn, TensorFlow, PyTorch), Golang, Java (Spring Boot), JavaScript/TypeScript, Node.js, React, Next.js, Vue.js, Nuxt.js, C#, .NET, ASP.NET MVC

Data & ML: Machine Learning, Recommendation Systems, Personalization, Ranking, Churn Prediction, Customer Segmentation, Image Matching & Similarity, Data Mining, Deep Learning, MLOps

Databases & Storage: MongoDB, PostgreSQL, MySQL, SQL Server, Elasticsearch, Redis, HBase, CouchDB

Infrastructure & DevOps: Docker, Kubernetes, Jenkins, Drone CI, Spinnaker, CI/CD Pipelines, Nginx, Varnish, Apache

Video & Streaming: HLS, MPEG-DASH, Smooth Streaming, DRM (PlayReady, FairPlay, Widevine), Video.js, Shaka Player, ExoPlayer

EDUCATION

Bachelor of Science in Physics Istanbul University | Istanbul, Turkey | 2000 - 2005 *Graduation Thesis: Chaos Theory / Predicting Weather Conditions*

PROFESSIONAL CERTIFICATIONS

Dynamic Adaptive Streaming Over HTTP Using MPEG-DASH Perpetual Solutions | 2014

Project Management and PMP Preparation Yildiz Technical University | 2012

C-SAP Secure Software Development Lostar | 2009

TRAININGS

- AWS Solution Architect (2018)
 - Deep Learning (2018)
 - Hadoop (2018)
 - Blockchain Programming (2018)
 - Product Management (2017)
 - Docker / Containers (2017)
 - Kubernetes (2017)
 - Microservice Fundamentals (2017)
 - Machine Learning (2013)
 - iPad and iPhone Application Development (2011)
 - iPhone Application Development (2010)
 - CMMI (2007)
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PORTFOLIO PROJECTS

GUI for CNC Machines (PLC Controller, CAD/CAM)

2013 - Present

Designed and implemented Visual C# user interfaces for CNC machine control systems for Schtec Machine.

Technologies: Visual C#, SQLite, TwinCAT, CAD/CAM

Multi-Language eCommerce Platform

2009 - Present

Architected and built complete eCommerce framework and platform (www.shoetekfiyat.com).

Technologies: ASP.NET MVC (C#), SQL Server, NHibernate, SQL Server Reporting Services, Bootstrap, LESS, jQuery

Pronet iPhone Panic Button Application

2013

Developed native iOS safety application (Awarded Altın Örümcek - Golden Spider Award).

Technologies: Xcode, Objective-C

Online Poker Hand Replayer

2012

Built interactive hand replay visualization for online poker platforms.

Technologies: ActionScript 3

Microsoft CRM and Navision Add-ons for Windows Mobile

2009

Developed mobile extensions for Microsoft CRM and Navision ERP systems.

Technologies: Visual C#

PROFESSIONAL READING

Selected influential books on leadership, engineering management, and technical excellence:

- Accelerate - Nicole Forsgren PhD
 - Team Topologies - Matthew Skelton and Manuel Pais
 - Multipliers - Liz Wiseman
 - Radical Candor - Kim Scott
 - Effective Executive - Peter Drucker
 - Sprint - Jake Knapp
 - Agile Testing - Janet Gregory
 - The Scrum Fieldbook - Jeff Sutherland
 - Objectives and Key Results - Paul R. Niven
 - Measure What Matters - John Doerr
 - The Manager's Path - Camille Fournier
 - Crucial Conversations - Kerry Patterson et al.
 - Data Mesh - Zhamak Dehghani
 - Collective Intelligence - Toby Segaran
 - Machine Learning: An Algorithmic Perspective - Stephen Marsland
 - Data Mining: Practical Machine Learning Tools and Techniques - Ian H. Witten, Eibe Frank, Mark A. Hall
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ADDITIONAL INFORMATION

Languages: Turkish (Native), English (Fluent) **Location:** Berlin, Germany **Work Authorization:** Permanent Residency

More about my leadership philosophy and working style: <https://hyurtseven81.github.io/>